Five emission calculations

One whole episode, one pollutant. O, S, R, M and A identify the existing calculations.

Notation and units. Bin i has speed v_i and count $d_i = q_i \Delta t$, with $\Delta t = 1/12$ h. $D = \sum_i d_i$; L is link length; C is hourly capacity; $T_f = L/v_f$. H denotes vehicle-hours, P episode duration and T per-vehicle travel time. $r_0(v)$ is the zero-acceleration lookup rate in g/(veh h); E is in grams. Counts and capacity use the same lane basis.

Shared fitted quantities for R, M and A.

$$x = \frac{D}{C}, \qquad \widehat{P} = \max(f_d x^n, x), \qquad \widehat{w}_{\max} = T_f f_p \widehat{P}^s, \qquad \widehat{\bar{w}} = \theta \widehat{w}_{\max}, \qquad H_{\text{pred}} = D(T_f + \widehat{\bar{w}}).$$

f_d, n, f_p, s, θ are calibrated by directional corridor and period, with the existing sparse-period fallback.

O Recorded-speed reference

Assign each bin's counted vehicles the complete-link travel time L/v_i, then sum:

$$H_i = d_i \frac{L}{v_i}, \qquad H_{\text{obs}} = \sum_i H_i, \qquad \boxed{E_O = \sum_i H_i r_0(v_i).}$$

The five-minute interval supplies the count; emission exposure is L/v_i, not five minutes.

S Episode mean-speed estimate

Use the count-weighted harmonic mean, preserving the reference's total vehicle-hours:

$$\bar{v}_{\text{obs}} = \frac{D}{\sum_i d_i/v_i} = \frac{DL}{H_{\text{obs}}}, \qquad \boxed{E_S = H_{\text{obs}}\, r_0(\bar{v}_{\text{obs}}).}$$

R Reconstructed whole-link speeds

For normalized entry time $u \in [0,1]$, define $w(u) = \widehat{w}_{\max} h(u)$ and $v_R(u) = L/[T_f + w(u)]$. The symmetric profile is

$$h(u) = \begin{cases} [4u(1-u)]^a, & \theta \le 2/3, \quad \int_0^1 h(u)\,du = \theta \text{ determines } a, \\ 1 - |2u-1|^b, & \theta > 2/3, \quad b = \theta/(1-\theta). \end{cases}$$

The entering-vehicle weight is $dN = D[1 + (\widehat{w}_{\max}/\widehat{P})h'(u)]\,du$. Thus

$$\boxed{E_R = \int_0^1 [T_f + w(u)]\, r_0\!\left(\frac{L}{T_f + w(u)}\right) dN.}$$

The integral uses 384 quadrature points; $\int dN = D$ and $\int [T_f + w(u)] dN = H_{\text{pred}}$. These are trip-average speeds by entry time, not individual operating-speed trajectories.

M Two-speed lookup

Infer the queue speed from discharge and the triangular fundamental diagram:

$$\widehat{\mu} = \frac{D}{\widehat{\bar{P}}}, \qquad \omega = \frac{C}{k_j - C/v_f}, \qquad \widehat{v}_q = \frac{\widehat{\mu}}{k_j - \widehat{\mu}/\omega}, \qquad k_j = 220 \text{ veh/mi/lane}.$$

Allocate mean travel time between two operating speeds while preserving distance:

$$\bar{T}_q = \frac{v_f \widehat{\bar{w}}}{v_f - \widehat{v}_q}, \qquad \bar{T}_{\text{free}} = T_f - \frac{\widehat{v}_q \widehat{\bar{w}}}{v_f - \widehat{v}_q}, \qquad \boxed{E_M = D[\bar{T}_{\text{free}} r_0(v_f) + \bar{T}_q r_0(\widehat{v}_q)].}$$

Here $\bar{T}_{\text{free}} + \bar{T}_q = T_f + \widehat{\bar{w}}$ and $v_f \bar{T}_{\text{free}} + \widehat{v}_q \bar{T}_q = L$.

A Two-speed cubic

Keep M's speeds and times; replace the lookup rate by its fitted cubic:

$$r_c(v) = c_0 + c_1 v + c_2 v^2 + c_3 v^3, \qquad \boxed{E_A = D[\bar{T}_{\text{free}} r_c(v_f) + \bar{T}_q r_c(\widehat{v}_q)].}$$

Evaluation: episode-total WAPE. For method $m \in \{S, R, M, A\}$ and the same $N = 137$ episodes,

$$\boxed{\text{WAPE}_m = 100 \frac{\sum_{k=1}^{N} |E_{m,k} - E_{O,k}|}{\sum_{k=1}^{N} E_{O,k}}.}$$

Absolute errors are calculated before pooling episodes. O is a detector-speed-based calculation, not measured emissions or observed vehicle trajectories. All five calculations omit acceleration effects.

Arizona detector 141: selected comparison

I-10 westbound. Constructed November 2016 average-weekday profile.

Why this case was selected

Among the 137 common episodes, four meet a strict diagnostic screen: absolute mean-trip-time error at most 5%; mean-delay, peak-delay, duration and inferred FD queue-speed errors each at most 10%; and the observed-derived finite-link check passes. This case has the largest CO_2 absolute-error gap between M and S among those four. These criteria select one diagnostic case; they do not estimate an overall method ranking.

Quantity	Observed-derived	Fitted	Difference (%)
Duration (min)	200.00	196.34	-1.83
Mean trip time (s)	117.97	112.80	-4.38
Mean delay (s)	59.55	54.38	-8.68
Peak delay (s)	118.80	125.05	+5.26
Queue FD speed (mph)	7.50	7.88	+5.07

Both observed-derived and fitted finite-link checks pass. Traffic predictions remain close under the stated five error screens, yet the emission estimates differ substantially. The FD queue speed is an inferred operating state; it is not the minimum recorded whole-link speed.

Episode-total emissions per lane

Method	CO_2 (kg)	NO_x (g)	CO (g)	HC (g)
O	616.895	558.764	5,435.873	228.367
S	666.081	633.654	6,280.692	247.442
R	658.749	660.958	5,357.030	227.059
M	862.035	1,050.327	6,265.620	261.263
A	846.162	1,016.092	5,928.745	255.425

Absolute percentage difference from O (%)

Method	CO_2	NO_x	CO	HC
O	Reference	Reference	Reference	Reference
S	7.97	13.40	15.54	8.35
R	6.78	18.29	1.45	0.57
M	39.74	87.97	15.26	14.40
A	37.16	81.85	9.07	11.85

For this single episode, the absolute percentage difference is $100|E_m - E_O|/E_O$. It is not a new pooled WAPE estimate.

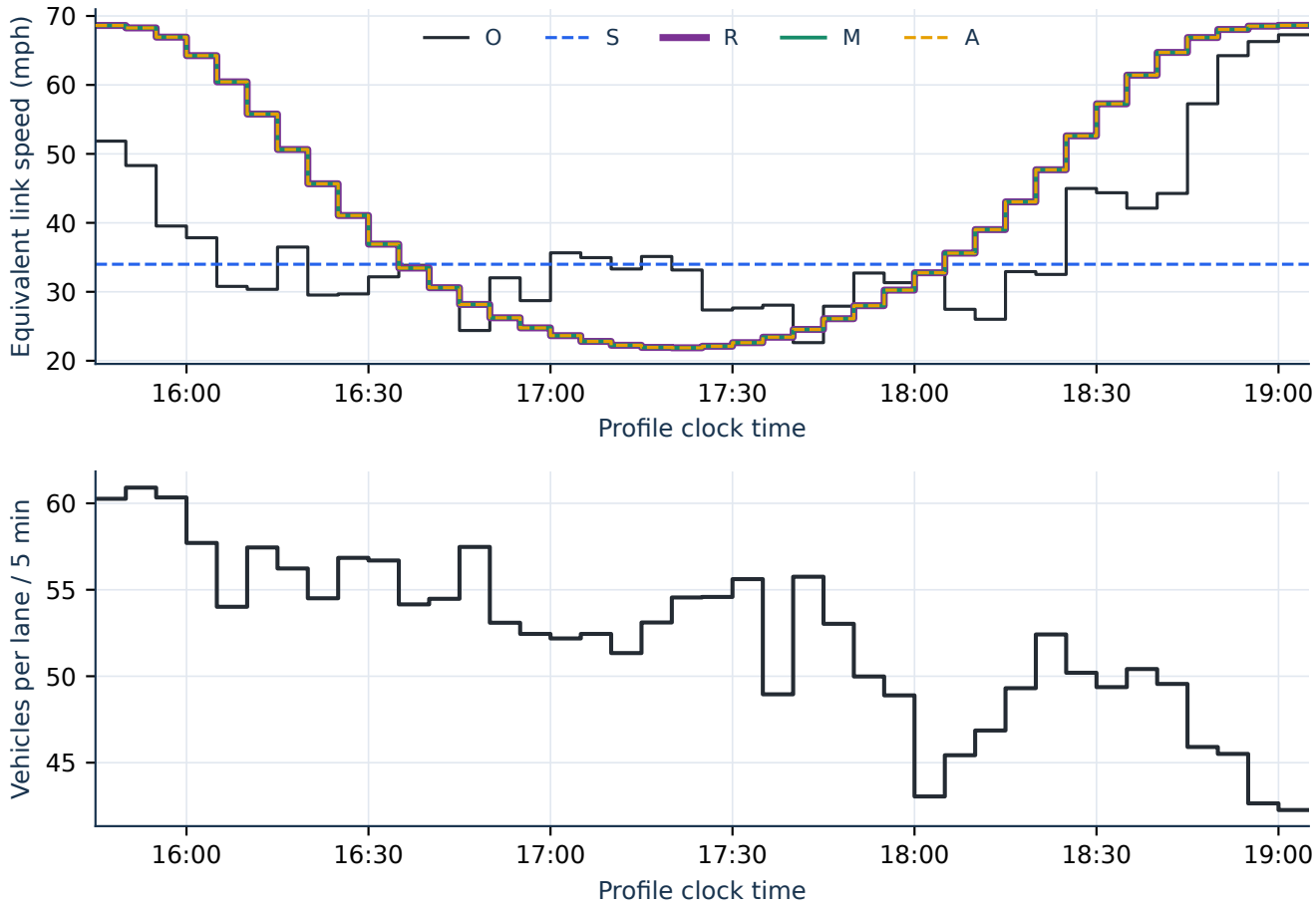
S uses the reference's observed vehicle-hours, while M uses a slightly smaller fitted exposure. Nevertheless, M is much higher than O. The following pages show how the time allocated to free-flow and queue operating speeds changes the effective emission rate. The reference contains no measured trajectories or exhaust.

Traffic inputs and speed profiles

One whole episode: 15:45 inclusive to 19:05 exclusive, 40 five-minute bins.

Input	Value	Input	Value
D (veh/lane)	2,089.97	L (mi)	1.114
C (veh/h/lane)	947.62	v_f (mph)	68.642
$x = D/C$ (h)	2.2055	T_f (s)	58.43
f_d; n	1.6932; 0.8330	f_p; s	0.3081; 1.6351
θ; k_j	0.4348; 220	Speed CV	0.2912

The AZ I-10 westbound PM fit uses 38 source episodes. This November profile averages 13 complete weekdays. The archived monthly profile is used directly; its date label identifies the month, not a single observed traffic day. The free-flow and FD card are detector-specific.



Equivalent speed for M/A. The time allocation gives

$$v_i^{\text{eff}} = \frac{v_f \bar{t}_{f,i} + v_q \bar{t}_{q,i}}{\bar{t}_{f,i} + \bar{t}_{q,i}} = \frac{L}{\bar{t}_{f,i} + \bar{t}_{q,i}}.$$

R, M and A overlap because they have identical allocated vehicle-hours and distance. Their operating speeds and rates differ. M/A use $v_f = 68.642$ and $v_q = 7.882$ mph internally, rather than the displayed equivalent speed.

The common clock is a display allocation: modeled vehicle groups are split into the observed five-minute count groups in cumulative-count order. It preserves every method's total. It does not infer when exhaust was physically released on the roadway.

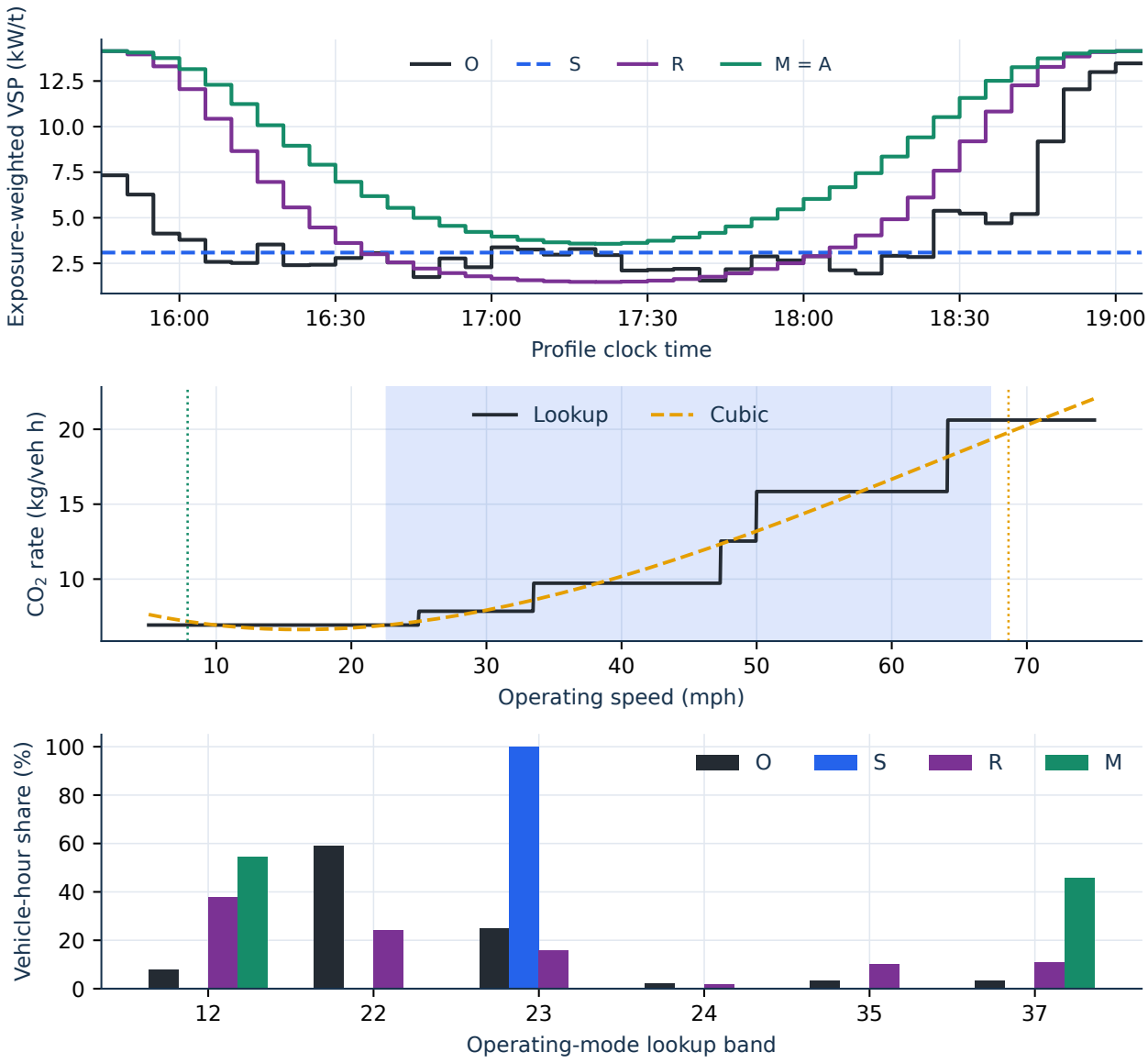
VSP profiles over the episode

Same count-aligned clock as the speed profiles; zero acceleration and level road.

Let $z(v) = \mathrm{VSP}_0(v)$ in kW/t and a_{ij} be the model-node count assigned to observed bin i. The plotted VSP is weighted by vehicle-hours:

$$\bar{z}_{R,i} = \frac{\sum_j a_{ij} T_j z(v_{R,j})}{H_{R,i}}, \qquad \bar{z}_{M,i} = \bar{z}_{A,i} = \frac{\sum_j a_{ij}[t_{f,j} z(v_f) + t_{q,j} z(v_q)]}{H_{R,i}}.$$

O uses $z(v_i)$; S uses the constant $z(\bar{v}_{\mathrm{obs}})$. M and A overlap because their operating speeds and exposure are identical. A changes only the emission-rate approximation. The supplied mapping is $z(v) = (0.156461 s_v + 0.00200193 s_v^2 + 0.000492646 s_v^3)/1.4788$, with $s_v = 0.44704 v$ for speed v in mph.



Top: VSP over time. Each step spans one observed five-minute bin. This exposure-weighted quantity generally differs from $z(v_i^{\mathrm{eff}})$, which is supplied separately in the CSV. It is not an emission rate or a measured trajectory VSP.

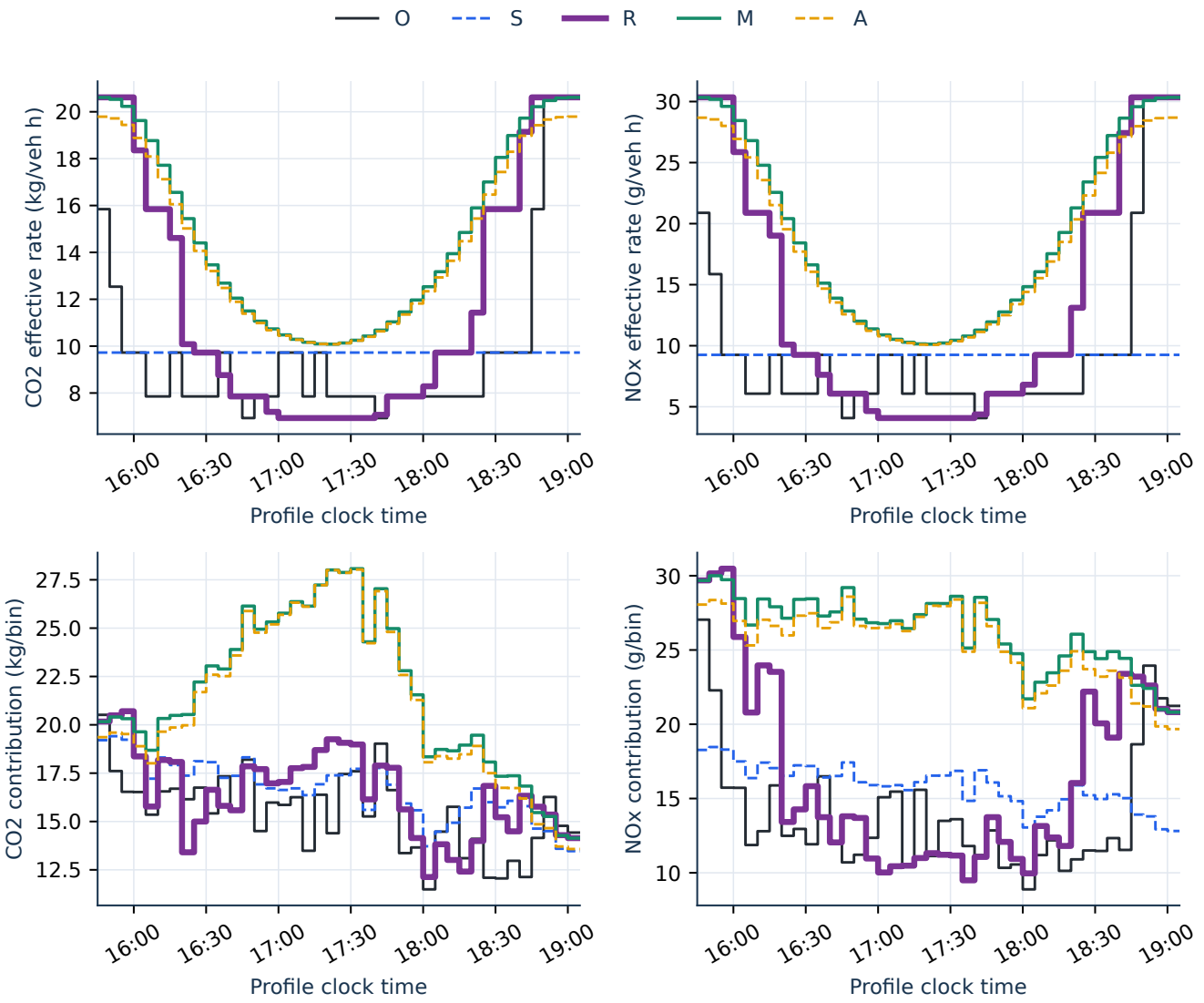
Middle: rate assignment. Shading marks observed speeds; dotted lines mark v_q (green) and v_f (orange). The lookup and cubic functions map the operating conditions to rates.

Bottom: total exposure by lookup band. O and R span several bands; S uses one band. M allocates exposure to modes 12 and 37. A uses those same two speeds with cubic rates.

Equal equivalent speed does not imply equal operating-state VSP. R, M and A share fitted vehicle-hours, but M/A can place substantial exposure at free-flow speed while compensating with a slow queue state. The VSP profile is a diagnostic average of those states, not a new rate input. The emission calculations and all 20 episode totals are unchanged.

Rates and five-minute contributions

Same observed count in each bin; method-specific travel-time exposure and rates.



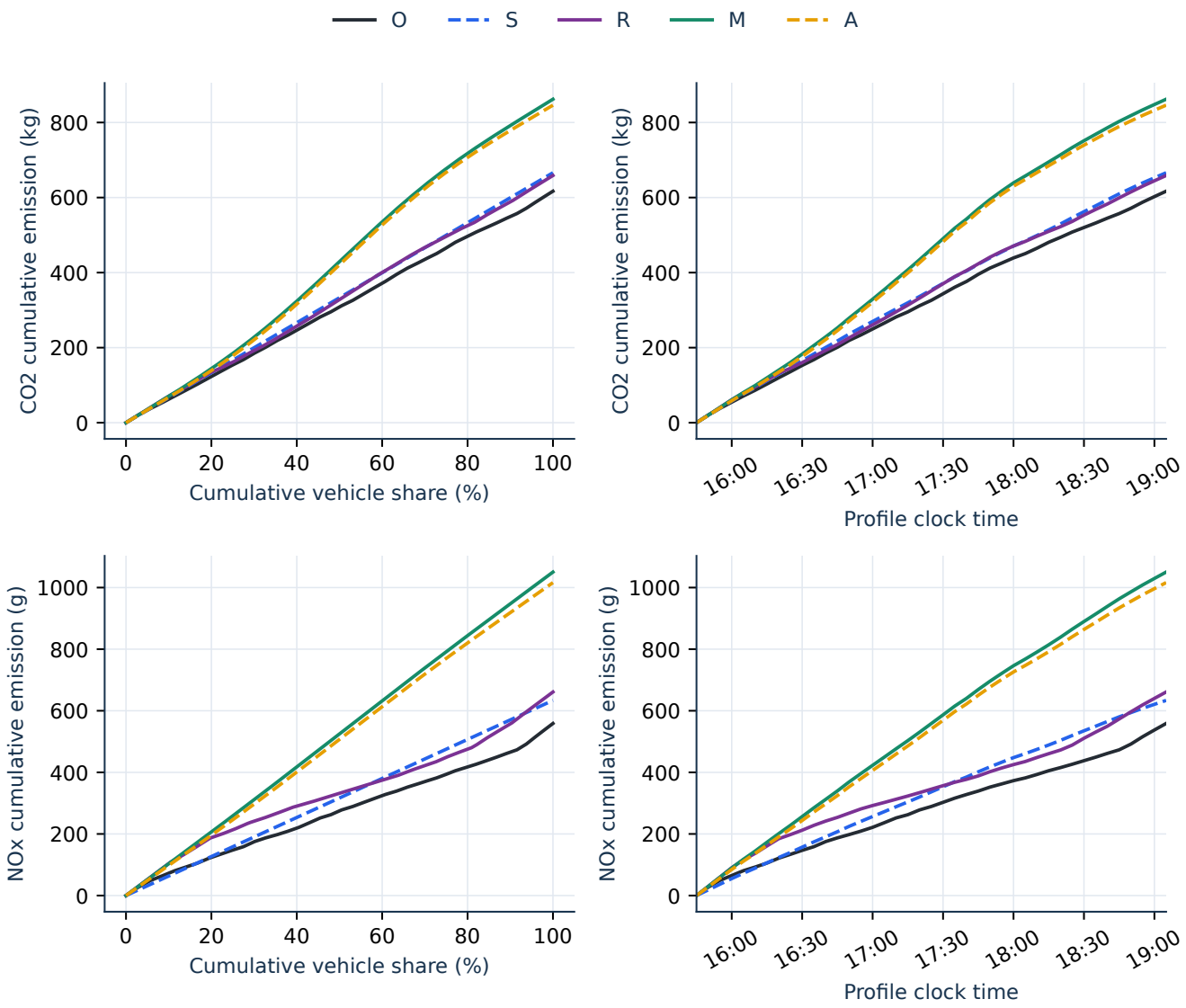
Top: effective rate. In each bin, $r^{\text{eff}}_{m,i} = E_{m,i}/H_{m,i}$. For R this averages the assigned rates of its reconstructed vehicle groups. For M/A it averages the free-flow and queue rates by their time allocations. It is generally not the lookup at the displayed equivalent speed.

Bottom: contribution from that bin's vehicles. $E_{m,i} = H_{m,i} r^{\text{eff}}_{m,i}$. The five-minute duration creates the vehicle count; a vehicle's exposure is its complete-link travel time. Each plotted series sums to the corresponding episode total on page 2.

O and S have identical total exposure. Their five-minute contributions differ because S assigns one constant trip time and rate to every counted vehicle. R/M/A share fitted exposure; their remaining differences come from the speeds at which rates are evaluated and from lookup versus cubic approximation.

How the contributions accumulate

Vehicle-order view on the left; conservative common-clock allocation on the right.



Vehicle share. O follows the observed bin order. R/M/A follow reconstructed entry-group order. S has constant emission per vehicle. All curves terminate at the same count D, with their own episode emission total.

Clock time. Each boundary value sums the complete-link emissions assigned to all preceding bins. Connecting lines interpolate bin contributions for display and imply no sub-bin exhaust timing.

For observed cumulative counts B_i and modeled cumulative counts N_j, the overlap count is

$$n_{ij} = \max\{0, \min(B_i, N_j) - \max(B_{i-1}, N_{j-1})\}.$$

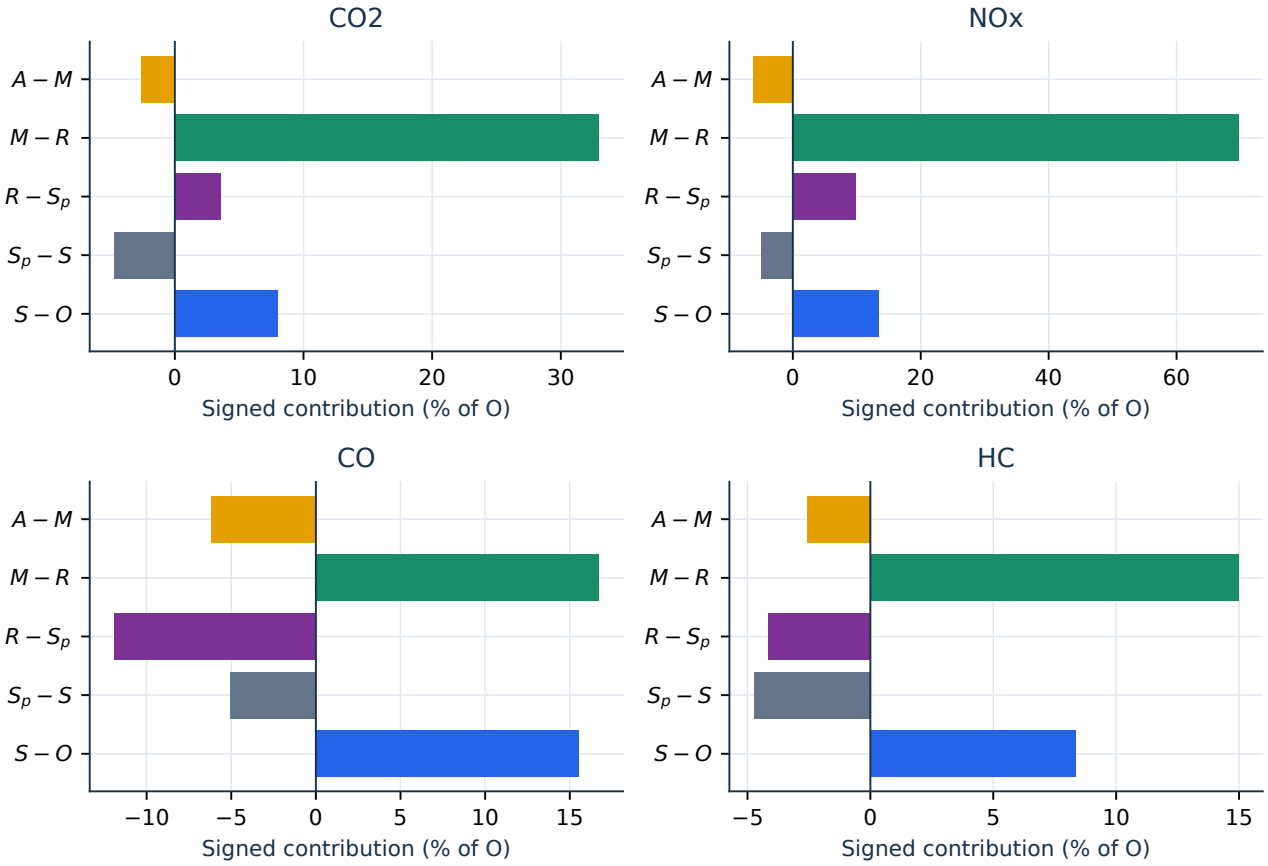
Use $H_i = \sum_j n_{ij} T_j$ and $E_i = \sum_j n_{ij} e_j$. Both count and emission totals are conserved. The underlying 384-node integration and all 40 common-clock rows are supplied as CSV files.

Where the differences enter

An exact signed accounting sequence, conditional on the chosen order.

Define $E_{S,p} = H_{\text{pred}} r_0(DL/H_{\text{pred}})$, a single-speed calculation with the fitted exposure shared by R/M/A. Then

$$E_A - E_O = (E_S - E_O) + (E_{S,p} - E_S) + (E_R - E_{S,p}) + (E_M - E_R) + (E_A - E_M).$$



Signed components as a percentage of O

Pollutant	S−O	S_p−S	R−S_p	M−R	A−M
CO_2	+7.97	-4.73	+3.54	+32.95	-2.57
NO_x	+13.40	-4.97	+9.86	+69.68	-6.13
CO	+15.54	-5.06	-11.93	+16.71	-6.20
HC	+8.35	-4.75	-4.18	+14.98	-2.56

The fitted exposure differs from O by -4.382%. $S_p - S$ combines this exposure change with any lookup-rate change caused by the fitted mean speed. $R - S_p$ adds reconstructed trip-speed variation. $M - R$ changes the operating-speed representation at equal fitted vehicle-hours. $A - M$ changes only the rate approximation.

The traffic screens pass, so a large travel-time calibration error is not needed to create this discrepancy. M and R have the same fitted vehicle-hours, but evaluate rates at different speed distributions. The large M–R component isolates that representation difference within this accounting sequence. It does not identify the unknown real trajectory emissions.

Audit and source files. All 20 episode totals reproduce the frozen matched-episode record. The supplied selection table contains all 137 cases, the five traffic error screens, the physical check and the four passing cases. The full-precision inputs, rate-band exposure, common-clock data, error components, editable vector figures and this report source are included with the case. No calibration, episode boundary or published result was replaced.